

Substitutability, Nonlinear Jacobi, and Sinkhorn as a Special Case

A short pedagogical note

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Abstract

Sinkhorn's algorithm is usually introduced as a variational algorithm: alternating Bregman projections, or coordinate ascent/descent on the dual of entropically regularized optimal transport. There is another point of view, emphasized in recent work around Galichon's equilibrium-flow program: Sinkhorn is a special case of a nonlinear Jacobi market-clearing algorithm. This second point of view does not require the equilibrium equations to be the gradient of a convex potential. The relevant structure is substitutability, formalized through *Z-functions*, *M-functions*, and *M₀-functions*. A familiar linear sufficient condition is a nonsingular M-matrix condition; in sign-uncertain formulations this is often expressed through an H-matrix condition. A worked lossy-transport example illustrates how the same Sinkhorn-like scaling formulas can remain convergent even when no convex OT potential exists.

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1 The basic fixed-point problem

Let Z be a finite set of coordinates, and let

$$Q : D \subseteq \mathbb{R}^Z \longrightarrow \mathbb{R}^Z$$

be a continuous map. In economic applications, $p \in \mathbb{R}^Z$ is a price vector and $Q_z(p)$ is excess supply in market z . The equilibrium problem is

$$Q(p) = 0. \tag{1}$$

The nonlinear Jacobi update clears each coordinate separately, holding the other coordinates fixed at their old values. Formally, for $p^t \in D$, define $p^{t+1} = T(p^t)$ by

$$Q_z(p_z^{t+1}, p_{-z}^t) = 0, \quad z \in Z, \quad (2)$$

where p_{-z} denotes all coordinates except z .

Assumption 1.1 (Responsiveness). For every z and every p_{-z} , the one-dimensional equation

$$Q_z(\pi, p_{-z}) = 0$$

has at least one solution in π . When there is more than one solution, we select the smallest one:

$$T_z(p) = \min \{ \pi : Q_z(\pi, p_{-z}) = 0 \}.$$

Remark 1.2 (Jacobi versus Gauss–Seidel). Equation (2) is the parallel Jacobi version. The familiar implementation of Sinkhorn often uses a Gauss–Seidel order: update rows using the old columns, then update columns using the new rows. Both are coordinate-clearing schemes; the order-theoretic hypotheses below are most transparent for Jacobi, and analogous monotone arguments apply to ordered coordinate updates.

2 Subsolutions and the first monotonicity argument

Definition 2.1 (Subsolution and supersolution). A vector p is a subsolution if $Q(p) \leq 0$ componentwise. It is a supersolution if $Q(p) \geq 0$.

The intended interpretation is that subsolution prices are too low and supersolution prices are too high. For this interpretation to be valid, own-price effects and cross-price effects need the right signs.

Definition 2.2 (Diagonal isotonicity and Z-functions). The map Q is diagonally isotone if, for each z , $Q_z(p_z, p_{-z})$ is nondecreasing in p_z .

It is a Z-function if, for each z , $Q_z(p_z, p_{-z})$ is antitone in p_{-z} : whenever $p_{-z} \leq p'_{-z}$,

$$Q_z(p_z, p_{-z}) \geq Q_z(p_z, p'_{-z}).$$

For C^1 maps, this means

$$\frac{\partial Q_z}{\partial p_{z'}}(p) \leq 0, \quad z' \neq z.$$

Proposition 2.3 (Monotone preservation of subsolutions). Assume Q is diagonally isotone and a Z-function. If p^t is a subsolution, then

$$p^t \leq p^{t+1} \quad \text{and} \quad p^{t+1} \text{ is again a subsolution.}$$

The analogous statement holds for supersolutions, with inequalities reversed.

Proof. Since p^t is a subsolution,

$$Q_z(p_z^t, p_{-z}^t) \leq 0 = Q_z(p_z^{t+1}, p_{-z}^t).$$

By diagonal isotonicity, $p_z^t \leq p_z^{t+1}$ for every z . Hence $p^t \leq p^{t+1}$. Now, using the Z-property,

$$Q_z(p_z^{t+1}, p_{-z}^{t+1}) \leq Q_z(p_z^{t+1}, p_{-z}^t) = 0.$$

Thus p^{t+1} is a subsolution. □

The preceding proposition is useful but not sufficient for convergence.

Example 2.4 (Why Z-structure alone is not enough). Let

$$Q(p) = \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} p.$$

Then Q is diagonally isotone and a Z-function, but the Jacobi update is

$$p_1^{t+1} = 2p_2^t, \quad p_2^{t+1} = 2p_1^t.$$

Except from the fixed point $p = 0$, the iterates diverge. The problem is that cross-effects dominate own-effects.

3 Nonreversingness, M-functions, and inverse isotonicity

The missing condition says that an increase in all prices cannot make all excess supplies fall. This is the nonlinear counterpart of the linear M-matrix idea.

Definition 3.1 (Nonreversingness). Let $Q : D \rightarrow \mathbb{R}^Z$.

(i) Q is strongly nonreversing if

$$p \geq p', \quad Q(p) \leq Q(p') \implies p = p'.$$

(ii) Q is weakly nonreversing if

$$p \geq p', \quad Q(p) \leq Q(p') \implies Q(p) = Q(p').$$

Definition 3.2 (M- and M_0 -functions). An M-function is a Z-function that is strongly nonreversing. An M_0 -function is a Z-function that is weakly nonreversing.

Thus an M-function has two ingredients:

- (a) substitutes: increasing other prices lowers excess supply for a given good;
- (b) own-price dominance: substitution effects cannot reverse the direction of the total price movement.

Theorem 3.3 (Inverse isotonicity). *Let Q be a Z-function. Then:*

(i) Q is an M-function if and only if it is inverse isotone:

$$Q(p) \leq Q(p') \implies p \leq p'.$$

(ii) Q is an M_0 -function if and only if its inverse is isotone as a set-valued map. In the single-valued notation, a convenient equivalent statement is

$$Q(p) \leq Q(p') \implies Q(p \wedge p') = Q(p), \quad Q(p \vee p') = Q(p'),$$

where $p \wedge p'$ and $p \vee p'$ denote componentwise minimum and maximum.

This theorem is the order-theoretic reason the algorithm works. It replaces convexity of a potential by monotonicity of the inverse equilibrium map.

4 Convergence of nonlinear Jacobi

Theorem 4.1 (Global monotone convergence for M-functions). *Assume Q is continuous, responsive, diagonally isotone, and an M-function. Assume also that there exist a subsolution $\underline{p}$ and a supersolution $\bar{p}$. Then the equation $Q(p) = 0$ has a unique solution p^* , and Jacobi iterates starting from any subsolution or any supersolution converge monotonically to p^* .*

Proof sketch. Starting from a subsolution, Proposition 2.3 gives an increasing sequence of subsolutions. Starting from a supersolution gives a decreasing sequence of supersolutions. By inverse isotonicity, every subsolution lies below every supersolution: indeed, if $Q(\underline{p}) \leq 0 \leq Q(\bar{p})$, then Theorem 3.3 yields $\underline{p} \leq \bar{p}$. Hence the lower and upper Jacobi sequences are monotone and bounded. Their limits solve $Q(p) = 0$ by continuity and responsiveness. If two solutions p, p' existed, then both $Q(p) \leq Q(p')$ and $Q(p') \leq Q(p)$; inverse isotonicity gives $p \leq p'$ and $p' \leq p$, hence $p = p'$. $\square$

Theorem 4.2 (The M_0 case). *If Q is an M_0 -function, uniqueness may fail. Under the same continuity, responsiveness, and lower-upper solution hypotheses, the solution set is a lattice. Jacobi started from the meet or the join of a subsolution and a supersolution converges to an extremal solution.*

Remark 4.3 (Why M_0 matters for Sinkhorn). Balanced optimal transport is invariant under adding the same constant to all dual potentials of one sign convention. Therefore the price-clearing map is typically M_0 , not M , before a normalization is imposed. Fixing one potential, or imposing a gauge such as $\sum p_z = 0$, removes the null direction and turns the reduced system into an M -system under the usual connectivity assumptions.

5 A differentiable M-matrix/H-matrix certificate

Suppose Q is C^1 on an order-convex domain, and write $J(p) = DQ(p)$.

Definition 5.1 (Z-, M-, and H-matrices). A matrix A is a Z-matrix if $A_{ij} \leq 0$ for $i \neq j$. A nonsingular M-matrix is a Z-matrix whose inverse is nonnegative; equivalently, $A = sI - B$ with $B \geq 0$ and $s > \rho(B)$.

The comparison matrix of A is

$$\langle A \rangle_{ii} = |A_{ii}|, \quad \langle A \rangle_{ij} = -|A_{ij}| \quad (i \neq j).$$

The matrix A is an H-matrix if $\langle A \rangle$ is an M-matrix. If A has positive diagonal entries, one often says A is an H_+ -matrix. For Z-matrices with positive diagonal, the H_+ and nonsingular M-matrix conditions coincide.

The following criterion is often the easiest way to recognize the condition informally remembered as an “H-matrix” condition.

Proposition 5.2 (Weighted diagonal dominance implies an M-function). *Assume Q is C^1 and satisfies*

$$\frac{\partial Q_i}{\partial p_j}(p) \leq 0, \quad i \neq j, \tag{3}$$

for all p . Suppose there exists a vector $\lambda \gg 0$ such that

$$\lambda^\top J(p) \gg 0 \quad \text{for every } p. \tag{4}$$

Equivalently, for each column j ,

$$\lambda_j \frac{\partial Q_j}{\partial p_j}(p) > \sum_{i \neq j} \lambda_i \left| \frac{\partial Q_i}{\partial p_j}(p) \right|. \tag{5}$$

Then Q is an M -function. If the strict inequalities are weakened to non-strict inequalities and $\lambda^\top J(p) \geq 0$, then Q is an M_0 -function.

Proof. Condition (3) says that Q is a Z -function. Let $p \geq p'$ and set $\delta = p - p' \geq 0$. If $\delta \neq 0$, then

$$\lambda^\top (Q(p) - Q(p')) = \int_0^1 \lambda^\top J(p' + t\delta) \delta dt > 0$$

by (4). Hence it is impossible to have $Q(p) \leq Q(p')$, because then the left side would be nonpositive. This proves strong nonreversingness. The weak version is the same argument with ≥ 0 : if $Q(p) - Q(p') \leq 0$ and its positive weighted sum is nonnegative, then the vector must be zero. $\square$

Remark 5.3 (Relation to H -matrices). When $J(p)$ is a Z -matrix, strict weighted diagonal dominance such as (5) implies that $J(p)$ is a nonsingular M -matrix, hence an H_+ -matrix. The H -matrix language is useful in numerical analysis because it survives some sign changes. The economic/substitution theory usually keeps the Z -sign pattern explicit and then speaks of M -functions.

6 Sinkhorn as the canonical special case

Let X and Y be finite sets. Let $a = (a_x)_{x \in X}$ and $b = (b_y)_{y \in Y}$ be positive masses with equal total mass, and let $K_{xy} > 0$ be a positive kernel. Entropically regularized optimal transport seeks a coupling π with row sums a and column sums b of the form

$$\pi_{xy} = r_x K_{xy} s_y, \quad r_x > 0, \quad s_y > 0. \quad (6)$$

The row and column constraints are

$$r_x \sum_y K_{xy} s_y = a_x, \quad s_y \sum_x K_{xy} r_x = b_y.$$

Sinkhorn's scaling updates are

$$r_x^{t+1} = \frac{a_x}{\sum_y K_{xy} s_y^t}, \quad s_y^{t+1} = \frac{b_y}{\sum_x K_{xy} r_x^t}. \quad (7)$$

The common alternating implementation uses the new r^{t+1} in the second equation; (7) is the parallel Jacobi version.

To put this into the M -function notation, set

$$u_x = \log r_x, \quad v_y = -\log s_y, \quad \pi_{xy}(u, v) = K_{xy} e^{u_x - v_y}.$$

Define $Q : \mathbb{R}^X \times \mathbb{R}^Y \rightarrow \mathbb{R}^X \times \mathbb{R}^Y$ by

$$Q_x(u, v) = \sum_{y \in Y} \pi_{xy}(u, v) - a_x, \quad (8)$$

$$Q_y(u, v) = b_y - \sum_{x \in X} \pi_{xy}(u, v). \quad (9)$$

Then $Q(u, v) = 0$ is exactly the row-column marginal system. The Jacobi equations $Q_x(u_x^{t+1}, v^t) = 0$ and $Q_y(u^t, v_y^{t+1}) = 0$ give

$$\begin{aligned} u_x^{t+1} &= \log a_x - \log \sum_y K_{xy} e^{-v_y^t}, \\ v_y^{t+1} &= \log \sum_x K_{xy} e^{u_x^t} - \log b_y. \end{aligned}$$

Exponentiating gives exactly (7).

The Z- and M_0 -structure

The Jacobian of (8)–(9) has entries

$$\begin{aligned}\frac{\partial Q_x}{\partial u_x} &= \sum_y \pi_{xy}, & \frac{\partial Q_x}{\partial v_y} &= -\pi_{xy}, \\ \frac{\partial Q_y}{\partial v_y} &= \sum_x \pi_{xy}, & \frac{\partial Q_y}{\partial u_x} &= -\pi_{xy}.\end{aligned}$$

All off-diagonal derivatives are nonpositive, so Q is a Z-function. Moreover, every column sum of $J = DQ$ is zero:

$$\mathbf{1}^\top J = 0.$$

Therefore the map is weakly nonreversing but not strongly nonreversing. The missing strong condition is exactly the additive gauge invariance

$$(u, v) \mapsto (u + c\mathbf{1}_X, v + c\mathbf{1}_Y),$$

under which $u_x - v_y$ and hence π_{xy} are unchanged. Thus balanced Sinkhorn is naturally an M_0 example. After fixing one potential or imposing a normalization, the reduced Jacobian is a weighted graph Laplacian minor; if the support graph of K is connected, that minor is a nonsingular M-matrix.

Why Sinkhorn is also variational

For entropic OT there is additionally a convex dual potential. In one sign convention,

$$\Phi(u, v) = \sum_{x,y} K_{xy} e^{u_x - v_y} - \sum_x a_x u_x + \sum_y b_y v_y,$$

up to irrelevant constants and sign choices. The first-order conditions are precisely $Q(u, v) = 0$. Sinkhorn can therefore be interpreted as coordinate descent/ascent on a convex objective. The order-theoretic point is different: the coordinate update only needs the market-clearing equations $Q_z = 0$ and the M- or M_0 -structure. It does not need Q to be a gradient.

7 A concrete OT-inspired non-variational example

The preceding Sinkhorn example is perfectly conservative: one unit leaving an origin arrives as one unit at a destination. Here is a small modification which keeps the same transport geometry and the same scaling form, but is generally not variational.

Let X and Y be finite sets, let $K_{xy} > 0$, and introduce positive outside-option coefficients $\sigma_x, \tau_y > 0$. We write

$$r_x = e^{u_x}, \quad s_y = e^{-v_y}, \quad \pi_{xy} = r_x K_{xy} s_y.$$

Think of π_{xy} as the physical amount shipped from x to y . Suppose, however, that only a fraction $\eta_{xy} \in (0, 1]$ is effective at the destination. The equilibrium equations are

$$\sigma_x r_x + \sum_{y \in Y} \pi_{xy} = a_x, \quad x \in X, \quad (10)$$

$$\tau_y s_y + \sum_{x \in X} \eta_{xy} \pi_{xy} = b_y, \quad y \in Y. \quad (11)$$

The first equation says that the mass available at origin x is split between the transport network and an outside sink. The second says that destination y can be served either by effective arrivals

or by a local outside option. This is not meant to be a standard OT model; it is a deliberately simple OT-shaped market-clearing system.

In the signed logarithmic variables (u, v) define

$$\begin{aligned} Q_x(u, v) &= \sigma_x e^{u_x} + \sum_y K_{xy} e^{u_x - v_y} - a_x, \\ Q_y(u, v) &= b_y - \tau_y e^{-v_y} - \sum_x \eta_{xy} K_{xy} e^{u_x - v_y}. \end{aligned}$$

Then $Q = 0$ is exactly (10)–(11). Clearing one coordinate at a time gives the Sinkhorn-like update

$$r_x^{t+1} = \frac{a_x}{\sigma_x + \sum_y K_{xy} s_y^t}, \quad s_y^{t+1} = \frac{b_y}{\tau_y + \sum_x \eta_{xy} K_{xy} r_x^t}. \quad (12)$$

Thus the algorithm still alternately rescales rows and columns. The only difference from Sinkhorn is that the marginal equations contain outside terms and the column equation sees the distorted arrivals $\eta_{xy} \pi_{xy}$.

The M-matrix certificate

The Jacobian has the same Z-sign pattern as Sinkhorn:

$$\begin{aligned} \frac{\partial Q_x}{\partial u_x} &= \sigma_x r_x + \sum_y \pi_{xy}, & \frac{\partial Q_x}{\partial v_y} &= -\pi_{xy}, \\ \frac{\partial Q_y}{\partial u_x} &= -\eta_{xy} \pi_{xy}, & \frac{\partial Q_y}{\partial v_y} &= \tau_y s_y + \sum_x \eta_{xy} \pi_{xy}. \end{aligned}$$

Assume $0 < \eta_{xy} \leq 1$ and

$$\tau_y > \sum_x (1 - \eta_{xy}) K_{xy} \frac{a_x}{\sigma_x}, \quad y \in Y. \quad (13)$$

After one update, $r_x \leq a_x / \sigma_x$. On the invariant domain where these inequalities hold, the vector $\lambda = \mathbf{1}$ certifies weighted diagonal dominance. Indeed, the column sum corresponding to u_x is

$$\sigma_x r_x + \sum_y (1 - \eta_{xy}) \pi_{xy} > 0,$$

while the column sum corresponding to v_y is

$$\tau_y s_y - \sum_x (1 - \eta_{xy}) \pi_{xy} = s_y \left[\tau_y - \sum_x (1 - \eta_{xy}) K_{xy} r_x \right] > 0$$

by (13). Hence $\mathbf{1}^\top DQ \gg 0$ on the invariant domain. By Proposition 5.2, this is an M-function there, so the coordinate-clearing iteration converges monotonically once initialized from a subsolution or supersolution in that domain.

The system is generally not variational. If $Q = \nabla \Phi$ for a C^2 potential Φ , then the cross-partials would have to agree:

$$\frac{\partial Q_x}{\partial v_y} = \frac{\partial Q_y}{\partial u_x}.$$

Here this would require $-\pi_{xy} = -\eta_{xy} \pi_{xy}$ for every active arc, hence $\eta_{xy} = 1$. Losses with $\eta_{xy} \neq 1$ therefore break exactness of the one-form $\sum_z Q_z dp_z$, even though the M-structure and the Sinkhorn-shaped update remain.

Numerical instance

Take two origins and two destinations,

$$K = \begin{pmatrix} 1 & e^{-1} \\ e^{-1} & 1 \end{pmatrix}, \quad a = b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \sigma = \tau = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

with loss factors

$$\eta = \begin{pmatrix} 0.80 & 0.90 \\ 0.70 & 0.85 \end{pmatrix}.$$

Condition (13) is immediate:

$$0.20 + 0.30e^{-1} \simeq 0.310 < 1, \quad 0.10e^{-1} + 0.15 \simeq 0.187 < 1.$$

Starting from $r^0 = s^0 = (1, 1)$, the parallel update (12) gives

t	r_1^t	r_2^t	s_1^t	s_2^t
0	1.0000	1.0000	1.0000	1.0000
1	0.4223	0.4223	0.4860	0.4585
2	0.6043	0.6108	0.6913	0.6672
3	0.5163	0.5204	0.6095	0.5817
6	0.5391	0.5441	0.6388	0.6119

The limit is approximately

$$r^* = (0.5371, 0.5421), \quad s^* = (0.6372, 0.6103).$$

The corresponding transport matrix is

$$\pi^* \simeq \begin{pmatrix} 0.3423 & 0.1206 \\ 0.1271 & 0.3308 \end{pmatrix}.$$

One can check the clearing equations directly:

$$\begin{aligned} r_1^* + \pi_{11}^* + \pi_{12}^* &\simeq 1, & r_2^* + \pi_{21}^* + \pi_{22}^* &\simeq 1, \\ s_1^* + 0.80\pi_{11}^* + 0.70\pi_{21}^* &\simeq 1, & s_2^* + 0.90\pi_{12}^* + 0.85\pi_{22}^* &\simeq 1. \end{aligned}$$

This is the closest finite-dimensional analogue of Sinkhorn in the note: positive kernel, multiplicative scalings, marginal-clearing equations, and closed-form row/column updates. The reason it converges, however, is the M-function condition, not a convex OT potential.

8 A minimal algebraic non-variational example

Let $a, b > 0$ and $0 \leq \lambda, \mu < 1$. Consider

$$\begin{aligned} Q_1(p_1, p_2) &= e^{p_1} - \lambda e^{p_2} - a, \\ Q_2(p_1, p_2) &= e^{p_2} - \mu e^{p_1} - b. \end{aligned}$$

Then

$$J(p) = \begin{pmatrix} e^{p_1} & -\lambda e^{p_2} \\ -\mu e^{p_1} & e^{p_2} \end{pmatrix}, \quad \mathbf{1}^\top J(p) = ((1 - \mu)e^{p_1}, (1 - \lambda)e^{p_2}) \gg 0.$$

Thus Proposition 5.2 applies: Q is an M-function. However, unless the cross-partials happen to agree everywhere,

$$\frac{\partial Q_1}{\partial p_2} = -\lambda e^{p_2} \neq -\mu e^{p_1} = \frac{\partial Q_2}{\partial p_1},$$

so Q is not the gradient of any scalar potential. The Jacobi update is nevertheless simple and convergent:

$$p_1^{t+1} = \log(a + \lambda e^{p_2^t}), \quad p_2^{t+1} = \log(b + \mu e^{p_1^t}).$$

The unique solution is obtained by solving the linear equations in $x = e^{p_1}$ and $y = e^{p_2}$:

$$x = a + \lambda y, \quad y = b + \mu x,$$

so

$$x^* = \frac{a + \lambda b}{1 - \lambda \mu}, \quad y^* = \frac{b + \mu a}{1 - \lambda \mu}.$$

This example is deliberately elementary, but it isolates the important point: the Jacobi/Sinkhorn-type update can converge for structural reasons that are not variational.

9 Equilibrium flows and imperfect transferability

The broader Galichon–Samuelson–Vernet framework replaces the additive OT inequality

$$p_x - p_y \geq -c_{xy}$$

by a nonlinear arc condition

$$p_x \geq G_{xy}(p_y),$$

where G_{xy} is increasing. An equilibrium flow consists of a node price vector p , nonnegative arc flows μ_{xy} , and net node flows q such that

$$\begin{aligned} \nabla^\top \mu &= q, \\ p_x &\geq G_{xy}(p_y), \quad xy \in A, \\ \mu_{xy} > 0 &\implies p_x = G_{xy}(p_y). \end{aligned}$$

The additive case $G_{xy}(p_y) = p_y - c_{xy}$ is the usual min-cost-flow/transport duality. Nonlinear G_{xy} accommodates time-dependent routing, nonlinear taxes, and imperfectly transferable utility in matching. In these cases, there need not be a convex objective whose gradient is the equilibrium equation. What remains is substitutability: increasing the relevant price at one node makes neighboring opportunities less attractive in the appropriate signed coordinates. The resulting equilibrium correspondence has inverse-isotone/lattice structure, and coordinate-clearing price updates are the natural generalization of Sinkhorn.

10 Takeaway

Sinkhorn sits at the intersection of two structures:

- (1) a variational structure: entropic OT is a convex optimization problem;
- (2) an order structure: the marginal equations form an M_0 -system, and Sinkhorn is a coordinate-clearing Jacobi/Gauss–Seidel method.

Galichon’s more general viewpoint keeps the second structure and drops the first. The lossy-transport example in Section 7 makes this separation concrete: the update has the same multiplicative scaling form as Sinkhorn, but asymmetric effective arrivals destroy the cross-partial symmetry required by a potential. The matrix condition to remember is best described as a nonlinear M-function condition. In smooth finite-dimensional problems, a practical certificate is a Z-Jacobian plus weighted diagonal dominance, equivalently an M-matrix or H_+ -matrix condition after the relevant sign convention and normalization.

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